

Data-assisted coordinated robust optimization method for urban water-power systems

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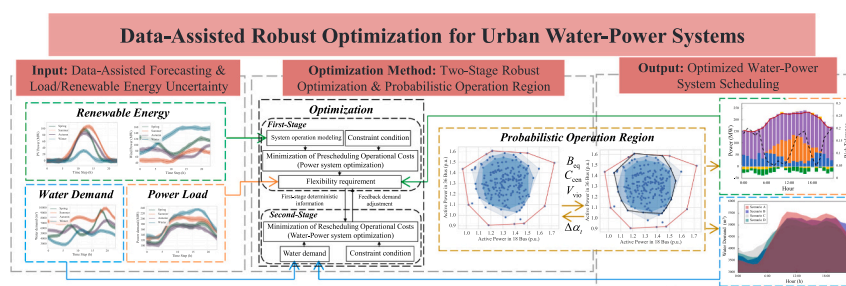
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HIGHLIGHTS

- Proposes a two-stage robust optimization for water-energy systems, coordinating water storage and energy dispatch.
- Introduces a dynamic risk-tolerance adjustment mechanism, enabling flexible control of model conservatism.
- Develops a probabilistic operation region to manage system uncertainties and support adaptive decisions.
- Establishes an evaluation system for the probabilistic operation region, providing key metrics for system robustness.

GRAPHICAL ABSTRACT



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ABSTRACT

With the increasing power demand of water distribution systems, utilizing pumps and water tanks as flexibility resources for the power system can effectively enhance the operational flexibility and efficiency of integrated energy systems. This paper proposes a data-assisted two-stage robust optimization method, incorporating a dynamic risk tolerance adjustment mechanism, to coordinate the operation of power and water supply systems, improve renewable energy penetration, and reduce overall operational costs. The proposed method first employs data-assisted techniques to generate uncertainty predictions and applies two-stage robust optimization to determine safe operational boundaries. Then, a convex hull method is used to construct probabilistic operation regions under different scheduling risk tolerances. Simultaneously, a dynamic risk tolerance adjustment mechanism is introduced, where the risk tolerance parameter is iteratively updated based on real-time evaluation indices to enhance the system's adaptability to uncertainties. Experimental results demonstrate that the proposed method effectively coordinates power-water system operations, enhances the flexibility of the water distribution system, improves renewable energy utilization, and reduces peak loads and reserve capacity requirements in the power system. Furthermore, the dynamic risk tolerance adjustment mechanism and probabilistic operation regions enrich scheduling strategies, enabling the system to dynamically optimize scheduling schemes based on load fluctuations and renewable energy forecasting accuracy, achieving an optimal balance between security and economy.

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1. Introduction

With the global energy system transitioning towards low-carbon development, the share of renewable energy in the electricity mix is increasing rapidly. The World Energy Outlook 2023 predicts that by 2030, renewable energy will account for nearly 50 % of total electricity production [1]. However, due to the intermittency and uncertainty of wind and solar power, power systems face greater scheduling challenges, making the role of flexibility resources increasingly critical [2]. Insufficient flexibility may lead to wind and solar curtailment, resulting in economic losses and limiting the integration of renewable energy [3]. Against this backdrop, water systems have emerged as important resources for enhancing grid flexibility due to their dispatchability and energy storage potential [4]–[5]. Water tanks and pumps in urban water distribution systems not only consume electricity but can also be optimally scheduled to balance grid load, reduce peak power demand, and improve adaptability to renewable energy [6]. Therefore, developing an optimization scheduling model that coordinates the operation of water distribution and power systems is crucial for improving grid stability, reducing operational costs, and increasing the utilization of renewable energy.

In recent years, as the role of water systems in power system flexibility management has gained widespread attention, an increasing number of studies have focused on exploring water-power coordinated optimization methods to enhance renewable energy integration and optimize operational costs. Reference [7] proposed an optimization approach based on chance constraints and stochastic programming, integrating machine learning-based forecasting models to optimize water-power-renewable energy hybrid scheduling and improve system flexibility. Reference [8] developed an optimization model that coordinates second-level frequency response (FRF) and 15-min power regulation flexibility (PRF) to enhance stability in a wind-hydro hybrid power system. Reference [9] further incorporated probabilistic distribution methods to assess the impact of renewable energy output and load forecasting errors, thereby improving grid security. Reference [10] introduced a two-stage nonlinear mixed-integer optimization model, coordinating wind power uncertainty using nonlinear programming and second-order cone programming (SOCP). Reference [11] adopted a joint probabilistic constraint optimization method, leveraging piecewise linearization techniques to transform the non-convex optimization problem in water-power systems into a solvable mixed-integer linear programming (MILP) problem for optimizing power and water supply costs. Reference [12] employed a distributionally robust optimization strategy combined with the Benders decomposition method to address false data injection attacks (FDIA) and renewable energy uncertainties. Reference [13] proposed a two-stage stochastic co-optimization approach that couples the power system with a reverse osmosis seawater desalination plant (RO-WDP) to reduce costs and operational risks. Reference [14] utilized second-order cone programming (SOCP) to analyze the cascading effects of power outages on water distribution systems, identifying critical vulnerable nodes and proposing optimization strategies. Reference [15] introduced a water-energy co-optimization method based on big data analytics to mitigate operational risks in a 100 % renewable energy scenario. Reference [16] employed a quasi-convex hull approach to transform a mixed-integer nonlinear programming (MINLP) problem into a mixed-integer convex programming (MICP) problem, thereby improving optimization computational efficiency.

However, existing studies primarily focus on source-side optimization in integrated water-power systems, while research on optimizing the scheduling of urban water distribution systems as demand-side resources within the demand-side management (DSM) framework remains relatively limited [17]. Additionally, current optimization approaches often concentrate on individual subsystems without fully considering the interactions between water and power networks, making it difficult to achieve globally optimal scheduling [18]. Although previous studies

have explored the co-optimization of water and power systems [19], several limitations remain, including single-objective optimization, a lack of uncertainty modeling, and limited applications of data-driven techniques. Additionally, due to the complex structure of urban water networks and the uncertainty of water demand, directly utilizing water tanks as energy storage may lead to water supply challenges or additional costs.

To address these issues, this paper proposes a data-assisted two-stage robust optimization method with dynamic risk tolerance adjustment to effectively coordinate the optimized operation of urban water networks and power systems. By integrating data-driven uncertainty forecasting, robust optimization, and dynamic risk tolerance adjustment, the proposed method optimizes water pump scheduling while enhancing renewable energy penetration, ensuring water supply reliability, and adaptively balancing system security and economy under different risk tolerance levels. The main contributions of this paper are as follows:

1. Propose a data-assisted water-power system coordination optimization method, which integrates data-driven uncertainty forecasting techniques to incorporate the stochastic characteristics of water and power loads into the scheduling optimization process, enabling more precise joint optimization modeling.
2. Develop a probabilistic operation region construction method and evaluation framework, which utilizes the convex hull method to establish operational boundaries under different scheduling schemes and introduces quantitative indicators to assess system flexibility and the adaptability of scheduling strategies under various uncertainty levels.
3. Introduce a dynamic risk tolerance adjustment mechanism within the robust optimization framework, which dynamically adjusts the risk tolerance parameter based on real-time system conditions, allowing the optimization scheme to flexibly control conservativeness and adaptively balance security and economy during the scheduling process.

The remainder is structured as follows: [Section 2](#) introduces the operational constraints of the integrated water-power system and the construction and evaluation of probabilistic operation regions. [Section 3](#) presents the solution methodology and simplification techniques. [Section 4](#) validates the feasibility and superior performance of the proposed method through case studies. [Section 5](#) provides conclusions and future research directions.

2. Operational and optimization methods for the integrated electricity-water energy system

Introducing water tanks and variable-speed pumps as flexibility resources in urban water networks can help the power system integrate more renewable energy and balance power loads. The power sector can utilize these facilities to adjust loads and meet the electricity demand of the water network, while the water sector can optimize scheduling to reduce operational costs or lease controllable assets to provide stable services. Independently optimizing the power and water systems may fail to achieve a globally optimal solution. Therefore, it is necessary to develop a coordinated scheduling strategy to ensure both power and water supply security, enhance renewable energy utilization, improve grid flexibility, and reduce overall operational costs.

2.1. Operational constraints and scheduling model

2.1.1. Power system

Eqs. (1) and (2) represent the active and reactive power balance conditions at the nodes. Eqs. (3) to (6) define the branch transmission capacity constraints. Eqs. (7) to (10) describe the generator unit states, as well as the start-up and shut-down operations. Eq. (11) specifies the operating conditions of the units. Eqs. (12) and (13) concern the

utilization of renewable energy and the range of load shedding. Eqs. (14) to (18) represent the value range and uncertainty range constraints of the power system.

$$P_i^G + P_i^R - P_i^D = \sum_{j \in \mathcal{N}} P_{ij}^L \quad (1)$$

$$Q_i^G - Q_i^D = \sum_{j \in \mathcal{N}} Q_{ij}^L \quad (2)$$

$$P_{ij}^L = V_i V_j (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij}) - G_{ij} V_i^2 \quad (3)$$

$$Q_{ij}^L = V_i V_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij}) + B_{ij} V_i^2 \quad (4)$$

$$(P_{ij}^L)^2 + (Q_{ij}^L)^2 = (S_{ij}^L)^2 \quad (5)$$

$$|S_{ij}| \leq S_{ij}^{\max} \quad (6)$$

$$O_{i,t} - O_{i,t-1} = S_{i,t} - E_{i,t} \quad (7)$$

$$S_{i,t} + E_{i,t} \leq 1 \quad (8)$$

$$S_{i,t} + \sum_{l=1}^{TS_i-1} E_{i,t+l} \leq 1 \quad (9)$$

$$E_{i,t} + \sum_{l=1}^{TE_i-1} S_{i,t+l} \leq 1 \quad (10)$$

$$\left| \sum_{l=1}^L P_{i,t,l}^G - \sum_{l=1}^L P_{i,t-l}^G \right| \leq K_i \quad (11)$$

$$P_{r,t}^R - P_{r,t}^C \geq 0 \quad (12)$$

$$P_{n,t}^D - P_{n,t}^C \geq 0 \quad (13)$$

$$(P_k^G)^{\min} \leq P_k^G \leq (P_k^G)^{\max} \quad (14)$$

$$(Q_k^G)^{\min} \leq Q_k^G \leq (Q_k^G)^{\max} \quad (15)$$

$$(P_{r,t}^R)^{\min} \leq P_{r,t}^R \leq (P_{r,t}^R)^{\max} \quad (16)$$

$$(P_{r,t}^D)^{\min} \leq P_{r,t}^D \leq (P_{r,t}^D)^{\max} \quad (17)$$

$$(V_i)^{\min} \leq V_i \leq (V_i)^{\max} \quad (18)$$

where $\mathcal{N}$ represents the set of nodes in the power system; P_i^G and Q_i^G denote the active and reactive power of the generator; P_i^R represents the active power from renewable energy sources; P_i^D and Q_i^D represent the active and reactive power loads; P_{ij}^L and Q_{ij}^L denote the active and reactive branch power flow from node i to node j ; P_i^C represents the power curtailment (either from renewable energy or load); V_i represents the voltage magnitude at node i ; G_{ij} and B_{ij} represent the conductance and susceptance of the branch; θ_{ij} is the voltage phase angle difference between the two nodes; S_{ij}^L represents the apparent power limit of branch (i,j) . The terms $O_{i,t}$, $S_{i,t}$, $E_{i,t}$ represent the operational status of generator unit i at time t (1 indicates operational, 0 indicates stopped); TS_i and TE_i represent the minimum up-time and down-time of generator unit i ; K_i denotes the power ramping threshold.

2.1.2. Water distribution system

Eqs. (19) to (22) describe the water flow and supply-demand balance conditions. Eqs. (23) to (25) define the relationship between the water

head and flow rate. Eqs. (26) to (29) represent the relationship between pump power and head: when the pump is operational, the flow rate satisfies the constraint $q_{\min} \leq q_{p,t} \leq q_{\max}$; when the pump is not operating, the flow rate is $q_{p,t} = 0$. Eqs. (30) and (31) describe the transmission and concentration variation equations for water pipes. Eqs. (32) to (39) represent the physical constraints of the water distribution system.

$$W_t^R - \gamma_t^W \cdot W_t^D = Q_t + \Lambda^T [E_t^I - E_t^O] \quad (19)$$

$$q_{m,t}^{\text{IN}} + q_{m,t}^{\text{OUT}} = (h_{m,t+1} - h_{m,t}) \cdot S^{\text{Tank}} \quad (20)$$

$$W_{m,t} = (1 - \mu_W) W_{m,t-1} + q_{m,t-1}^{\text{IN}} + q_{m,t-1}^{\text{OUT}} \quad (21)$$

$$W_{m,t} C_m - W_{m,t-\Delta t} C_{m,t-\Delta t} = \left(\sum q_{nm} C_{nm} - \sum q_{mk} C_m - k_b C_m \right) \cdot \Delta t \quad (22)$$

$$H_t = G \cdot \Pi(Q_t) |Q_t|^{1.852} \quad (23)$$

$$h_{m,t} - h_{n,t} = \zeta \cdot \text{sign}(q_{mn,t}) \cdot q_{mn,t}^2 \quad (24)$$

$$-M(1 - z_{mn,t}) \leq h_{n,t} - h_{m,t} + J_{mn,t} \leq M(1 - z_{mn,t}) \quad (25)$$

$$\Delta H_{p,t} = \Omega_t^2 \left[a - b \left(\frac{Q_t}{\Omega_t} \right)^c \right] \quad (26)$$

$$P_t^{\text{Pump}} = \frac{\rho g}{\eta_{\text{pump}}} \Delta H_{p,t} \cdot Q_{p,t} \quad (27)$$

$$\hat{h}_{p,t} = \Omega_{i,t}^2 (a q_{p,t}^2 + b q_{p,t} + c) \quad (28)$$

$$P_{i,t}^{\text{Pump}} = \Omega_{i,t}^3 (k_1 q_{p,i,t}^3 + k_2 q_{p,i,t}^2 + k_3 q_{p,i,t}) \quad (29)$$

$$\frac{\partial C_{mn}}{\partial t} + \frac{q_{mn}}{S_{mn}} \frac{\partial C_{mn}}{\partial x} + \tau C_{mn} = 0 \quad (30)$$

$$C_m \cdot \left(\sum q_{nm} + Q_{ex} \right) = \sum q_{nm} C_{nm} + Q_{ex} C_{ex} \quad (31)$$

$$0 \leq Q_t \leq (Q_t)^{\max} \quad (32)$$

$$(W_{m,t})^{\min} \leq W_{m,t} \leq (W_{m,t})^{\max} \quad (33)$$

$$0 \leq P_t^{\text{Pump}} \leq (P_t^{\text{Pump}})^{\max} \quad (34)$$

$$(H_t)^{\min} \leq H_t \leq (H_t)^{\max} \quad (35)$$

$$J_{mn,t} \geq 0 \quad (36)$$

$$0 \leq E_t^I \leq (E_t^I)^{\max} \quad (37)$$

$$0 \leq E_t^O \leq (E_t^O)^{\max} \quad (38)$$

$$0 \leq \gamma_t^W \leq 1 \quad (39)$$

where Q_t represents the water flow rate in the pipeline; Q_{ex} denotes external flow; C_{mn} is the concentration of substances in the pipeline; C_{ex} is the concentration of external fluid; W_t^R represents the amount of water injected into the water node from the tank; W_t^D represents the water demand at the water node; $W_{m,t}$ is the amount of water in the tank; Ω_t represents the pump speed; $q_{p,t}$ is the pump flow rate; ΔH_t^p denotes the head increment of the pump; P_t^{Pump} represents the pump power; η_{pump} is the pump efficiency; ζ is the proportional constant, and $\text{sign}(q_{mn,t})$ denotes the sign function of the flow rate; H_t represents the pressure head at the water node; $h_{m,t}$ and $h_{n,t}$ are the head heights at the two ends of the

pipeline, nodes m and n ; $J_{mn,t}$ represents the pressure drop across the pressure regulating valve; $z_{mn,t}$ is the state variable of the valve (0 for closed, 1 for open); G is the pipe friction coefficient; Λ is the pipe-to-node incidence matrix; E_t^i and E_t^o represent the inflow and outflow rates of the water tank; γ_t^w is the water supply ratio; ρ is the water density; τ is the reaction rate constant; g is the gravitational acceleration; a , b , and c are the pump parameters; S^{Tank} represents the cross-sectional area of the water tank; μ_w is the water loss coefficient; k_b is the reaction rate constant in the tank; and k_1, k_2, k_3 are the coefficients for pump power consumption.

2.2. Two-stage robust optimization method

Robust optimization aims to ensure system stability under the worst-case conditions. Two-stage robust optimization refines output constraints while achieving optimization objectives, reducing the impact of uncertainties and enhancing scheduling accuracy and reliability.

2.2.1. Objective function

The operating cost of the power system comprises generation costs and penalty terms associated with renewable energy curtailment. The primary objective of the function is to minimize both costs. For the water distribution system (WDS), the objective function incorporates the cost of power provision along with penalties for unmet demand. Consequently, the optimization of the WDS seeks to minimize these costs as well.

The objective function for the integrated system is as follows:

$$\mathcal{E} = \min_{\mathcal{X}, \mathcal{Z}} \left\{ C_1(\mathcal{X}) + \max_{\mathcal{U}} \left[\min_{\mathcal{Y}} C_2(\mathcal{Y}, \mathcal{U}) + C_3(\mathcal{Z}, \mathcal{U}) \right] \right\} \quad (40)$$

where $\mathcal{X}, \mathcal{Z}$ represent the first-stage decision variables; $\mathcal{Y}$ represents the second-stage decision variables, and $\mathcal{U}$ denotes the uncertainty variables.

C_1 represents the total operating cost of traditional generators, including startup and shutdown costs:

$$C_1(\mathcal{X}) = \sum_{i=1}^N \sum_{t=1}^T (c_i^O O_{i,t} + c_i^S S_{i,t} + c_i^E E_{i,t}) \quad (41)$$

where c_i^O, c_i^S, c_i^E represent the operation, startup, and shutdown costs of generator i .

C_2 represents the power system operating cost, which includes generation costs and penalties for renewable energy curtailment:

$$C_2(\mathcal{Y}, \mathcal{U}) = \eta_1 \cdot \sum_{i=1}^N \sum_{t=1}^T P_{i,t}^G + \sum_{i=1}^{N_g} \sum_{t=1}^T \sum_{l=1}^L K_{i,l} P_{i,t}^G \quad (42)$$

where $K_{i,l}$ is the cost coefficient for generator i under cost category l ; η_1 is the penalty coefficient.

C_3 represents the operating cost of the water distribution system:

$$C_3(\mathcal{Z}, \mathcal{U}) = c_w \sum_{m=1}^{N_w} \sum_{t=1}^T P_{m,t}^D + \eta_2 \cdot \sum_{t=1}^T \left(\sum_{m=1}^{N_w} P_{m,t}^W - \sum_{m=1}^{N_w} P_{m,t}^D + P_t^\Delta \right)^2 \quad (43)$$

where $P_{m,t}^W$ represents the power supply, provided by power generation P_i^G and renewable energy P_i^R ; $P_{m,t}^D$ represents the power demand, primarily consumed by water pumps; P_t^Δ represents the flexibility demand or residual power of the water distribution system; η_2 is the penalty coefficient.

In the WDS, pumps are employed to sustain water flow when pressure is insufficient, while valves regulate pressure or control water flow within the pipelines. Tanks function as reservoirs to store or supply water to the system. Pumps elevate pressure and transport water through the consumption of electricity, with greater electricity usage

enabling higher water flow rates. Tanks store water to maintain the balance between supply and demand.

From the perspective of the power system, the WDS can be regarded as a controllable load. By optimizing the operation of pumps and tanks, the WDS can fulfill the flexibility requirements of the power system and integrate more renewable energy.

2.2.2. Framework structure

The objective of the optimization is to minimize operating costs. To achieve this, the integrated power-water system optimization is carried out in three phases, as illustrated in Fig. 1. The first phase entails probabilistic forecasting of renewable energy and load uncertainties, where the accuracy of these forecasts directly impacts the subsequent optimization outcomes. The second phase optimizes power system demand, determining any surplus or deficit, which reflects the system's flexibility. The third phase focuses on optimizing the water distribution system's response to the power system's flexibility requirements while ensuring the WDS operates normally.

This paper adopts the Two-Stage Robust Optimization (TSRO) method, which considers uncertainty factors to optimize the coordinated operation of a high-renewable-energy power system and the WDS. System flexibility is quantified by the proportion of renewable energy and curtailment rates, with the goal of identifying the optimal operational strategy for the integrated system. Furthermore, the uncertainty set constructed more closely reflects real-world conditions, avoiding the need to divide into multiple typical scenarios. In contrast, fluctuations in water demand are relatively minor and, as such, are excluded from the calculations.

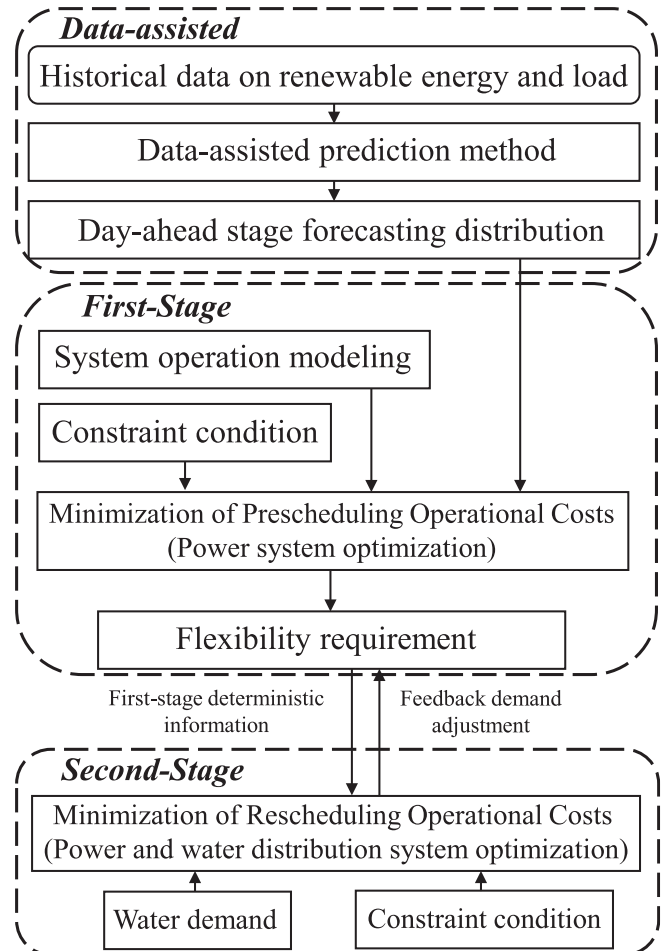


Fig. 1. Two-stage robust optimization scheduling framework.

2.3. Modeling and scheduling evaluation of the probabilistic operation region

2.3.1. Probabilistic operation region

This study proposes a coordinated scheduling model for the urban power grid and water distribution system and defines the secure operating region Ω for the integrated system. The decision space includes the power system's generation output P^G and the WDS's energy storage regulation E^W , which can be expressed as:

$$\Omega = \{y | f(x) = y, g(x, y) \leq 0\} \quad (44)$$

where $y = (P^G, E^W)$ represents the output variables, and $x = (V, \theta, \pi)$ includes voltage V , phase angle θ , and water distribution parameter π . The function $f(x)$ describes the nonlinear relationship of energy flow between the power and water systems, while the constraint $g(x, y) \leq 0$ ensures the secure operation constraints of both systems.

Boundary points correspond to conditions where specific constraints reach equality, while others still satisfy inequality constraints. To determine the feasible range of each variable in y , this study formulates and solves a nonlinear optimization problem:

$$\left(\underline{y}_n, \overline{y}_n \right) = \{ \min y_n, \max y_n \}, \quad n \in \mathcal{N} \quad (45)$$

By solving $\mathcal{N}$ optimization problems for each variable separately, the feasible range of y , denoted as $[\underline{y}, \overline{y}]$, can be determined. Subsequently, a convex hull approximation is employed to construct the operating region, as shown in Fig. 2. The convex hull is represented as:

$$\text{Conv}(\mathcal{Z}) = \left\{ z = \sum_{i=1}^{k_z} \delta_i z_i | z_i \in \mathcal{Z}, \delta_i \geq 0, \sum_{i=1}^{k_z} \delta_i = 1 \right\} \quad (46)$$

where k_z selects the boundary points of $\mathcal{Z}$ as convex hull vertices.

Due to the stochastic nature of renewable energy, a deterministic operating region Ω may fail to fully address uncertainty disturbances. To address this, this study introduces the concept of the Probabilistic Operation Region (POR), as illustrated in Fig. 3. The POR is defined as the region formed by all possible output combinations of the integrated system under uncertain renewable energy generation $\mathcal{Z}^R$:

$$\Omega^R = \{P_1^R, P_2^R, \dots, P_n^R\} \quad (47)$$

where each P_i^R represents a possible output combination.

The boundary points of POR satisfy the following constraints:

$$\begin{aligned} f(x, P^R) &= y \\ g(x, y, P^R) &\leq 0 \end{aligned} \quad (48)$$

By incorporating renewable energy uncertainty, the POR provides

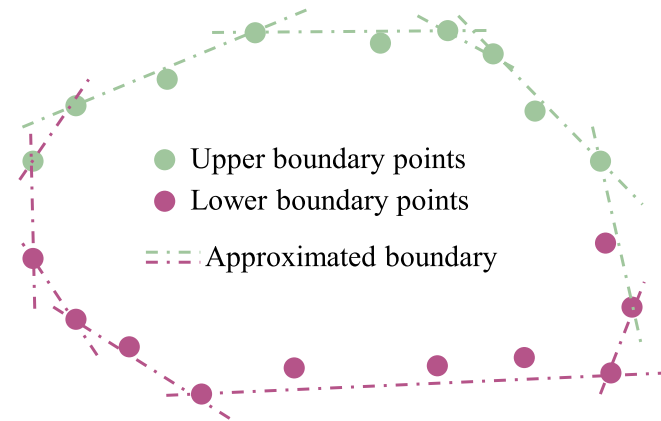


Fig. 2. Operational boundary constructed by convex hull.

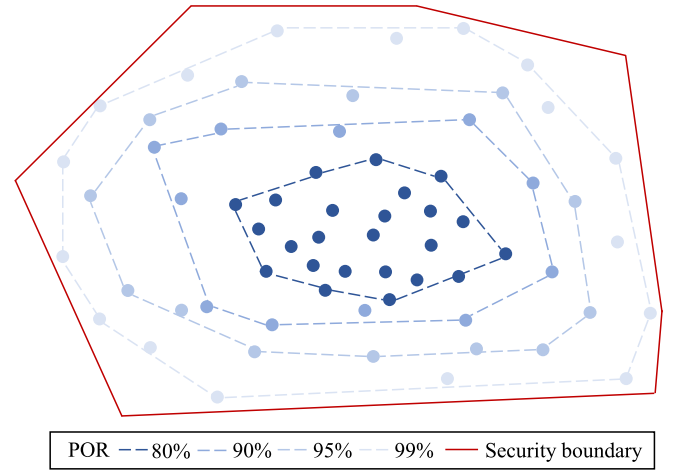


Fig. 3. Probabilistic operational region based on convex hull.

the feasible operational range and safety margins of the system. This approach allows for evaluating scheduling strategies under different risk tolerances, eliminating the need for repeated power flow calculations at each time step, thereby improving scheduling efficiency.

Under the uncertainty set $\mathcal{Z}^R$, if the feasible scheduling region $\mathcal{Z}^R$ is a convex set, then all solutions within Ω^R are feasible. The proof of this is provided in Appendix A.

2.3.2. Scheduling evaluation under uncertainty

Under the uncertainty optimization framework, the scheduling results are no longer a single unit dispatch solution but a distribution of possible outputs. To systematically evaluate the flexible range of generation units, their adjustment probabilities, and their impact on system security, this study proposes a POR evaluation index that considers uncertainty. This method can guide system scheduling, dynamically adjusting risk tolerance based on different operational states and risk preferences, thereby optimizing the scheduling strategy.

The Security Index B_{eq} quantifies how far an operational point is from the POR boundary. A higher value indicates that generation dispatch is closer to the secure central region:

$$B_{eq} = \frac{1}{X} \sum_{x=1}^X \frac{d_x}{d_{max}} \quad (49)$$

$$d_x = \min_{1 \leq j \leq m} \left\{ \frac{|(y_{j+1} - y_j)x_q - (x_{j+1} - x_j)y_q + x_{j+1}y_j - y_{j+1}x_j|}{\sqrt{(y_{j+1} - y_j)^2 + (x_{j+1} - x_j)^2}} \right\} \quad (50)$$

where d_x represents the minimum distance between the operational point $Q = (x_q, y_q)$ and the POR boundary, while $P_m = (x_m, y_m)$ denotes the boundary vertices of the POR.

The Compactness Index C_{cen} evaluates the distribution concentration of generation outputs. A lower value suggests less uncertainty in dispatch, meaning generation points are concentrated within a smaller range:

$$C_{cen} = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{M} \sum_{j=1}^M \sqrt{(x_{ij} - x_{cen})^2 + (y_{ij} - y_{cen})^2} \right) \quad (51)$$

where (x_{cen}, y_{cen}) is the geometric center, calculated as:

$$\begin{cases} x_{\text{cen}} = \frac{1}{m} \sum_{k=1}^m x_k \\ y_{\text{cen}} = \frac{1}{m} \sum_{k=1}^m y_k \end{cases} \quad (52)$$

The Violation Index V_{vio} quantifies the extent to which operational points deviate from the POR boundary, reflecting the feasibility of system operation under uncertainty:

$$V_{\text{vio}} = \frac{\sum_{i=1}^k w_i \cdot (N_{i,\text{out}} \cdot d_i)}{\sum_{i=1}^k w_i \cdot N_i} \quad (53)$$

$$d_i = \frac{d_{i,\text{avg}}}{d_{\text{max}}} \quad (54)$$

where $d_{i,\text{avg}}$ represents the average deviation in the i -th state.

2.3.3. Dynamic adjustment and adaptive optimization

In uncertainty optimization scheduling, the setting of risk tolerance α directly affects both the security and economy of the system. A lower value of α corresponds to a larger uncertainty set, providing a higher safety margin for the system but increasing the operational cost. A higher value of α allows a more compact operating region, improving economy, but it may reduce the system's robustness. To address this, this paper proposes a dynamic adjustment method for risk tolerance based on evaluation indices. The value of α is adjusted in real-time by assessing B_{eq} , C_{cen} , and V_{vio} , thereby achieving an adaptive balance between security and economy in the scheduling strategy.

The risk tolerance α is dynamically updated at each scheduling time step t by the following recursive formula:

$$\alpha_t = \alpha_{t-1} + \Delta\alpha_t \quad (55)$$

where the adjustment $\Delta\alpha_t$ is determined by the evaluation indices:

$$\Delta\alpha_t = k_1 \cdot (B_{\text{eq}} - B_A) - k_2 \cdot (C_{\text{cen}} - C_A) - k_3 \cdot V_{\text{vio}} \quad (56)$$

where B_A is the safety index threshold; C_A is the concentration index threshold; and k_1, k_2, k_3 are weight coefficients.

When B_{eq} falls below the threshold or V_{vio} is too high, the system's operating point is close to or exceeds the boundary of the POR, indicating insufficient safety margin. In this case, α should be reduced to expand the operating region and enhance conservatism. When C_{cen} is too high, the unit outputs are dispersed, leading to greater uncertainty, and α should be reduced to tighten the operating region, improving scheduling determinism. Conversely, if B_{eq} is too high, C_{cen} is too low, and V_{vio} is close to zero, the system is stable, and α can be moderately increased to shrink the operating region, optimizing economy. This dynamic adjustment mechanism ensures that risk tolerance α changes with the system's operational state, achieving a balance between security and economy in the scheduling strategy.

Based on the dynamic adjustment of risk tolerance α , an adaptive POR can be constructed to ensure the system's optimal operating state under different risk levels. This method includes the following steps:

1. Uncertainty adjustment: Based on the risk tolerance α , calculate and generate the corresponding probabilistic confidence intervals. A lower α corresponds to a larger uncertainty set, covering a wider range and ensuring a larger safety margin for the system, but with higher operational costs. A higher α forms a more compact operating region, improving economy, but may reduce the system's scheduling flexibility.
2. Boundary construction: At each α_i level, calculate the system's POR Ω^R . By using the convex hull method, the boundaries of the operating region are computed and approximated as the safe operating region Ω , ensuring system stability at different risk tolerance levels.

3. Evaluation and optimization: Calculate the current operating domain's B_{eq} , C_{cen} , and V_{vio} , and combine them with the dynamic adjustment strategy to optimize α , ensuring that the final selected operating region achieves a dynamic balance between security and economy.

3. Solution methodology

3.1. Data-assisted method

In time series forecasting tasks, in addition to considering correlations between periods, it is essential to account for the inherent periodicity of the data itself. In this study, the Hilbert-Huang Transform (HHT) is employed to extract the periodicity of one-dimensional time series data [20]. Consider a one-dimensional time series $\mathbf{X}_{1D} \in \mathbb{R}^{T \times C}$, where T represents the time length and C the channel dimension.

First, Empirical Mode Decomposition (EMD) is applied to decompose $\mathbf{X}_{1D}$ into several Intrinsic Mode Functions (IMFs). Then, the Hilbert Transform is performed on each IMF to obtain the instantaneous amplitude $\mathbf{A}$ and instantaneous frequency information for each IMF. The average amplitude $\mathbf{A}$ is defined as the mean of the instantaneous amplitudes of all IMFs:

$$\mathbf{A} = \text{Avg}(\text{Amp}(\text{HHT}(\mathbf{X}_{1D}))) \quad (57)$$

Next, the top k instantaneous frequencies with the highest amplitudes are selected, denoted as $\{f_1, f_2, \dots, f_k\}$. These frequencies correspond to the most prominent periodic components in the signal, represented as:

$$\{f_1, \dots, f_k\} = \underset{f \in \left\{1, \dots, \left\lceil \frac{T}{2} \right\rceil\right\}}{\text{argTopk}}(\mathbf{A}) \quad (58)$$

The period length corresponding to each selected frequency is then calculated, denoted as $\{p_1, p_2, \dots, p_k\}$, where p_i represents the period length corresponding to the selected frequency f_i . The period length is obtained by dividing the total length T by the frequency:

$$p_i = \left\lceil \frac{T}{f_i} \right\rceil \quad i \in \{1, 2, \dots, k\} \quad (59)$$

where $\lceil x \rceil$ denotes the smallest integer greater than or equal to x .

This process effectively reveals the periodic characteristics of the signal and facilitates the understanding and analysis of the time-frequency characteristics of the time series data. Based on the selected periods, the original one-dimensional time series $\mathbf{X}_{1D}$ is folded. This process can be formalized as:

$$\mathbf{X}_{2D}^i = \text{Reshape}_{p_i, f_i}(\text{Padding}(\mathbf{X}_{1D})), i \in \{1, \dots, k\} \quad (60)$$

where $\text{Padding}(\cdot)$ represents padding with zeros at the end of the sequence so that the sequence length becomes divisible by p_i .

Finally, a set of two-dimensional tensors $\{\mathbf{X}_{2D}^1, \mathbf{X}_{2D}^2, \dots, \mathbf{X}_{2D}^k\}$ is obtained, where each $\mathbf{X}_{2D}^i$ represents the two-dimensional temporal variations dominated by period p_i . The transformation process is illustrated in Fig. 4. In each tensor, the rows and columns correspond to adjacent time points and periods, respectively, exhibiting two-dimensional locality. This locality implies that adjacent elements have similar values or trends in variation, allowing feature extraction from the sequence data through convolution operations.

The forecasting model is composed of stacked Forecast Blocks, as shown in Fig. 5. The input sequence $\mathbf{X}_{1D} \in \mathbb{R}^{T \times C}$ with a length of T is first passed through an embedding layer to obtain deep features $\mathbf{X}_{1D}^0 \in \mathbb{R}^{T \times d_{\text{model}}}$. For the l -th layer Forecast Block, the input is $\mathbf{X}_{1D}^{l-1} \in \mathbb{R}^{T \times d_{\text{model}}}$, which is then used to extract two-dimensional temporal variations through 2D convolution:

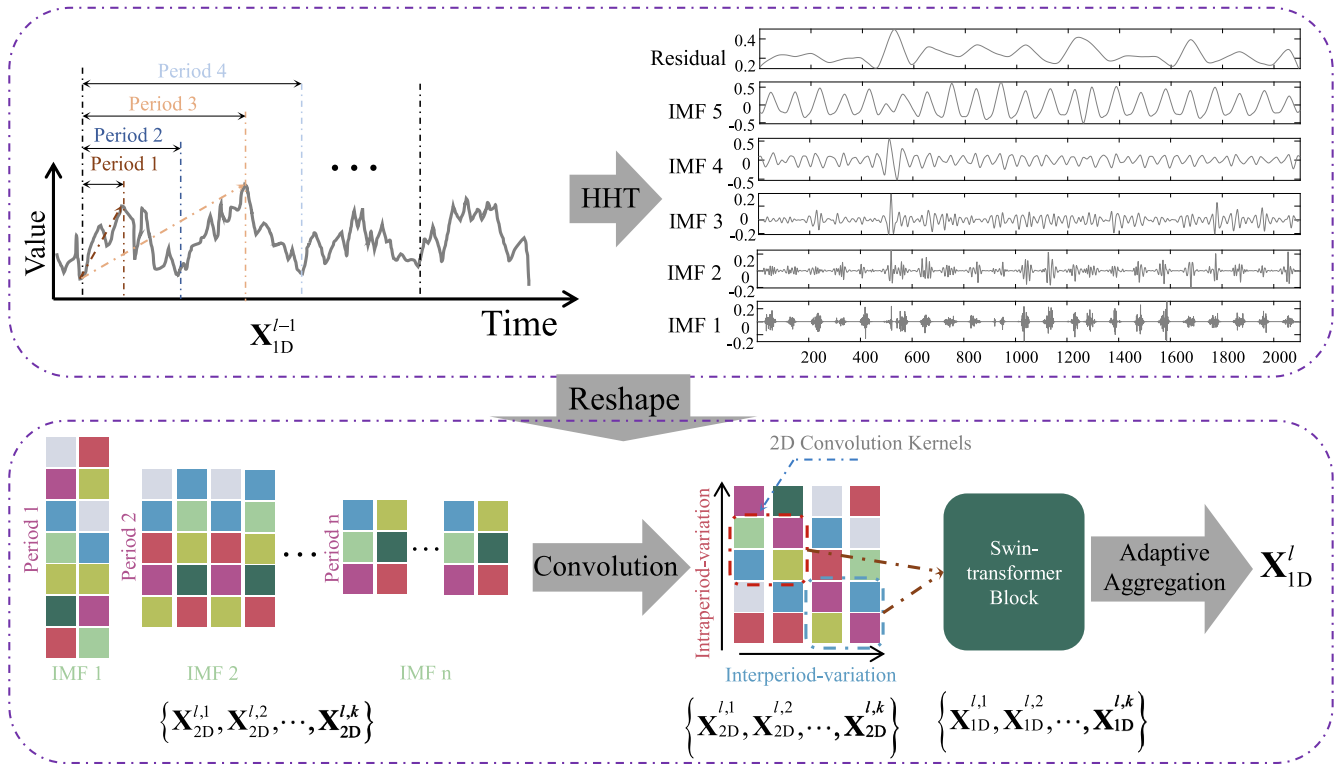


Fig. 4. The overall execution structure of the probability forecasting model.

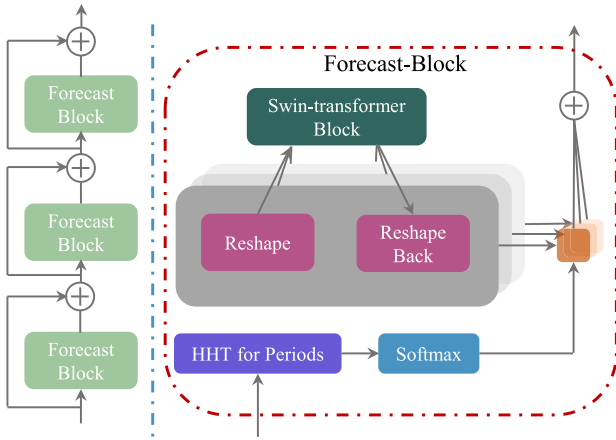


Fig. 5. The architecture of the forecast block.

$$X_{1D}^l = \text{Forecast-Block}(X_{1D}^{l-1}) + X_{1D}^{l-1} \quad (61)$$

Specifically, the Forecast Block consists of the following sub-processes:

1. Transformation from 1D to 2D: First, the periodic characteristics of the input one-dimensional time series features X_{1D}^{l-1} are extracted and converted into two-dimensional tensors to represent 2D temporal variations

$$A^{l-1}, \{f_1, \dots, f_k\}, \{p_1, \dots, p_k\} = \text{Period}(X_{1D}^{l-1}) \quad (62)$$

$$X_{2D}^{l,i} = \text{Reshape}_{p_i f_i}(\text{Padding}(X_{1D}^{l-1})), i \in \{1, \dots, k\} \quad (63)$$

2. Extraction of 2D Temporal Variation Features: For the two-dimensional tensors $\{X_{2D}^{l,1}, X_{2D}^{l,2}, \dots, X_{2D}^{l,k}\}$, given their 2D locality, 2D convolution is applied to extract features. In this work, the Swin

Transformer model [21] is used:

$$\hat{X}_{2D}^{l,i} = \text{Swin-transformer}(X_{2D}^{l,i}), i \in \{1, \dots, k\} \quad (64)$$

3. Transformation from 2D to 1D: The extracted temporal features are transformed back into the one-dimensional space for information aggregation

$$\hat{X}_{1D}^{l,i} = \text{Trunc}(\text{Reshape}_{1, (p_i \times f_i)}(\hat{X}_{2D}^{l,i})), i \in \{1, \dots, k\} \quad (65)$$

where $X_{2D}^{l,i} \in \mathbb{R}^{p_i \times f_i \times d_{\text{model}}}$ is the i -th transformed 2D tensor, and $\text{Trunc}(\cdot)$ denotes the removal of the zero-padding applied during the $\text{Padding}(\cdot)$ operation, truncating it back to the original length T .

4. Adaptive fusion: Following the design of autoformer [22], the obtained one-dimensional representations $\{\hat{X}_{1D}^{l,1}, \dots, \hat{X}_{1D}^{l,k}\}$ are weighted and summed based on the intensity of their corresponding frequencies to produce the final output

$$\hat{A}_{f_1}^{l-1}, \dots, \hat{A}_{f_k}^{l-1} = \text{Softmax}(A_{f_1}^{l-1}, \dots, A_{f_k}^{l-1}) \quad (66)$$

$$X_{1D}^l = \sum_{i=1}^k \hat{A}_{f_i}^{l-1} \times \hat{X}_{1D}^{l,i} \quad (67)$$

3.2. Simplified treatment of constraints

The integrated power-water system constructed is a non-convex nonlinear programming (NLP) problem. Since obtaining a global optimal solution is challenging, it is necessary to linearize the constraints to achieve a more accurate solution within a shorter computation time. To effectively handle the non-convex nonlinear constraints, various linearization techniques have been employed in this study.

3.2.1. Power transmission calculation

The power transmission calculations in eqs. (3)–(4) are linearized by

small-angle approximation and the introduction of a new variable $W_{ij} = V_i V_j$. Given that the phase angle difference θ_{ij} is small, it uses $\cos\theta_{ij} \approx 1$ and $\sin\theta_{ij} \approx \theta_{ij}$ to simplify the calculations, which results in the linearized eqs. (68)–(69). Additionally, the transmission power limit constraints (5)–(6) are modified by retaining the second-order cone constraints and introducing a binary variable $x \in \{0, 1\}$ to represent the open/closed status of the transmission line, yielding eq. (70).

$$P_{ij}^L \approx W_{ij}(G_{ij} + B_{ij}\theta_{ij}) - G_{ij}V_i^2 \quad (68)$$

$$Q_{ij}^L \approx W_{ij}(G_{ij}\theta_{ij} - B_{ij}) + B_{ij}V_i^2 \quad (69)$$

$$\sqrt{\left(P_{ij}^L\right)^2 + \left(Q_{ij}^L\right)^2} \leq x \cdot S_{ij}^{\max} \quad (70)$$

3.2.2. Power ramp rate constraints

To handle the power ramp rate constraints (11), an auxiliary variable $\xi_{i,t}$ is introduced to represent the absolute value of the difference between two power quantities, leading to new eqs. (71)–(73).

$$\sum_{l=1}^L P_{i,t,l}^G - \sum_{l=1}^L P_{i,t-1,l}^G \leq \xi_{i,t} \quad (71)$$

$$-\sum_{l=1}^L P_{i,t,l}^G + \sum_{l=1}^L P_{i,t-1,l}^G \leq \xi_{i,t} \quad (72)$$

$$\xi_{i,t} \leq K_i \quad (73)$$

3.2.3. Water flow and supply-demand balance equations

In the water flow and supply-demand balance equations, linearization is applied to eqs. (19) and (22) by introducing auxiliary variables. For example, in eq. (19), the term $\gamma_t^W \cdot W_t^D$ is replaced by Z_t , and in eq. (22), the terms $W_{m,t} \cdot C_m$ and $W_{m,t-\Delta t} \cdot C_{m,t-\Delta t}$ are replaced by $Z_{m,t}$ and $Z_{m,t-\Delta t}$, resulting in the linearized eqs. (74)–(75).

$$W_t^R - Z_t = Q_t + \Lambda^T [E_t^I - E_t^O] \quad (74)$$

$$Z_{m,t} - Z_{m,t-\Delta t} = \left(\sum q_{nm} C_{nm} - \sum q_{mk} C_m - k_b C_m \right) \cdot \Delta t \quad (75)$$

3.2.4. Head-flow relationship

The head-flow relationship equation is linearized by introducing auxiliary variables. For instance, eq. (23) is simplified using auxiliary variables, leading to eq. (76), and in eq. (24), the term $q_{mn,t}^2$ is replaced by $Y_{mn,t}$, resulting in eq. (77).

$$H_t \geq G \cdot \Pi(Q_t) \cdot \sqrt{Q_t} \quad (76)$$

$$h_{m,t} - h_{n,t} = \zeta \cdot Y_{mn,t} \quad (77)$$

3.2.5. Pump power and head constraints

The pump power and head constraints are linearized using first-order Taylor expansion and piecewise linearization. For example, the nonlinear term $(Q_t/\Omega_t)^c$ in eq. (26) is replaced by the auxiliary variable $y_t = Q_t/\Omega_t$, and y_t^c is linearized, resulting in eq. (78). Eq. (27) is approximated using a Maclaurin series, and the term $\Delta H_{P,t} \cdot Q_{P,t}$ is replaced by an auxiliary variable Z_t , leading to the second-order cone constraint (79). Additionally, eq. (28) uses the auxiliary variable $u_{P,t} = q_{P,t}^2$ to obtain eq. (80). The power terms in eq. (29) are linearized by introducing multiple variables, $x_1 = q_{P,t}^3$ and $x_2 = q_{P,t}^2$, ultimately leading to the multivariable linear combination in eq. (81).

$$\Delta H_{P,t} \geq \Omega_t^2 \cdot (a - b \cdot y_t^c) \quad (78)$$

$$P_t^{\text{pump}} \geq \frac{\rho g}{\eta_{\text{pump}}} Z_t \quad (79)$$

$$\hat{h}_{P,t} \geq \Omega_{t,t}^2 \cdot (a u_{P,t} + b q_{P,t} + c) \quad (80)$$

$$P_{i,t}^{\text{pump}} \geq \Omega_{i,t}^3 \cdot (k_1 x_1 + k_2 x_2 + k_3 x_3) \quad (81)$$

3.2.6. Pipe transmission and concentration equations

The linearization of pipe transmission and concentration equations focuses on eqs. (30) and (31). In eq. (30), an auxiliary variable Z_{mn} is introduced to represent $\left(q_{mn}/S_{mn}\right) \cdot \left(\partial C_{mn}/\partial x\right)$, resulting in eq. (82). In eq. (31), the term $C_m \cdot q_{nm}$ is replaced by the auxiliary variable Z_{nm} , yielding eq. (83).

$$\frac{C_{mn,t} - C_{mn,t-\Delta t}}{\Delta t} + z_{mn} + \tau C_{mn,t} = 0 \quad (82)$$

$$\sum Z_{nm} + z_{ex} = \sum q_{nm} C_{nm} + Q_{ex} C_{ex} \quad (83)$$

The nonlinear functions are linearized and relaxed into a Mixed-Integer Quadratically Constrained Programming (MIQCP) model, which is then directly solved using solvers such as GUROBI.

3.3. C&CG algorithm solving process

The optimization objective and framework of the system have been detailed in Section 2.2. Based on the objective function (40) and the constraints, the C&CG algorithm is used to describe the master-slave problem relationship in the two-stage robust optimization. The execution process is shown in Fig. 6.

1. Initialization: Initialize the uncertainty set $\mathcal{U}$ and solve the initial master problem.

2. Iteration Process:

a. Master Problem: Based on the current uncertainty set $\mathcal{U}^*$, solve the master problem to obtain the current solutions $\mathcal{X}^*$, $\mathcal{Z}$ and λ^* .

$$\begin{aligned} \min_{\mathcal{X}, \mathcal{Z}, \lambda} \quad & C_1(\mathcal{X}) + \lambda \\ \text{s.t.} \quad & F(\mathcal{X}, \mathcal{Z}) \leq D \\ & \lambda \geq C_2(\mathcal{X}^*, \mathcal{U}^*) + C_3(\mathcal{Z}^*, \mathcal{U}^*) \\ & G(\mathcal{X}, \mathcal{Y}^*) + H(\mathcal{Z}, \mathcal{U}^*) \leq R, \quad \forall \mathcal{U}^* \in \mathcal{S} \end{aligned} \quad (84)$$

In the master problem, the objective is to minimize the total cost of the power and water distribution systems by solving for the optimal first-stage decision variables $\mathcal{X}$ and $\mathcal{Z}$, along with the penalty parameter λ . Here, $\mathcal{Y}^*$ and $\mathcal{U}^*$ denote the optimal solutions obtained from the Slave Problem, representing the worst-case second-stage decisions and associated uncertainties.

b. Slave Problem: Using the solutions $\mathcal{X}^*$, $\mathcal{Z}$ from the master problem, solve the slave problem to find the new worst-case scenario $\mathcal{U}^*$.

$$\begin{aligned} \max_{\mathcal{Y}, \mathcal{U}} \min_{\mathcal{Y}} \quad & C_2(\mathcal{Y}, \mathcal{U}) + C_3(\mathcal{Z}, \mathcal{U}) \\ \text{s.t.} \quad & G(\mathcal{X}^*, \mathcal{Y}) + H(\mathcal{Z}, \mathcal{U}) \leq R \end{aligned} \quad (85)$$

In the slave problem, the uncertainty $\mathcal{U}$ is configured in the most adverse way. For this configuration, the second-stage decision variables are solved to minimize the operational costs of the power and WDS. Based on the first-stage decisions $\mathcal{X}^*$ and $\mathcal{Z}$ given by the master problem, the slave problem generates the worst-case costs and constraints, which are then used to guide the solution of the master problem.

c. Update: Add the new $\mathcal{U}^*$ to the uncertainty set, update the master problem, and continue solving.

3. Termination Condition: The algorithm terminates when the slave problem cannot find a new, worse scenario or when the objective value λ^* of the master problem no longer changes.

Through this master-slave framework, the C&CG algorithm can effectively approximate the global optimal solution step by step, minimizing the operational costs of the combined power and water distribution systems while ensuring system robustness.

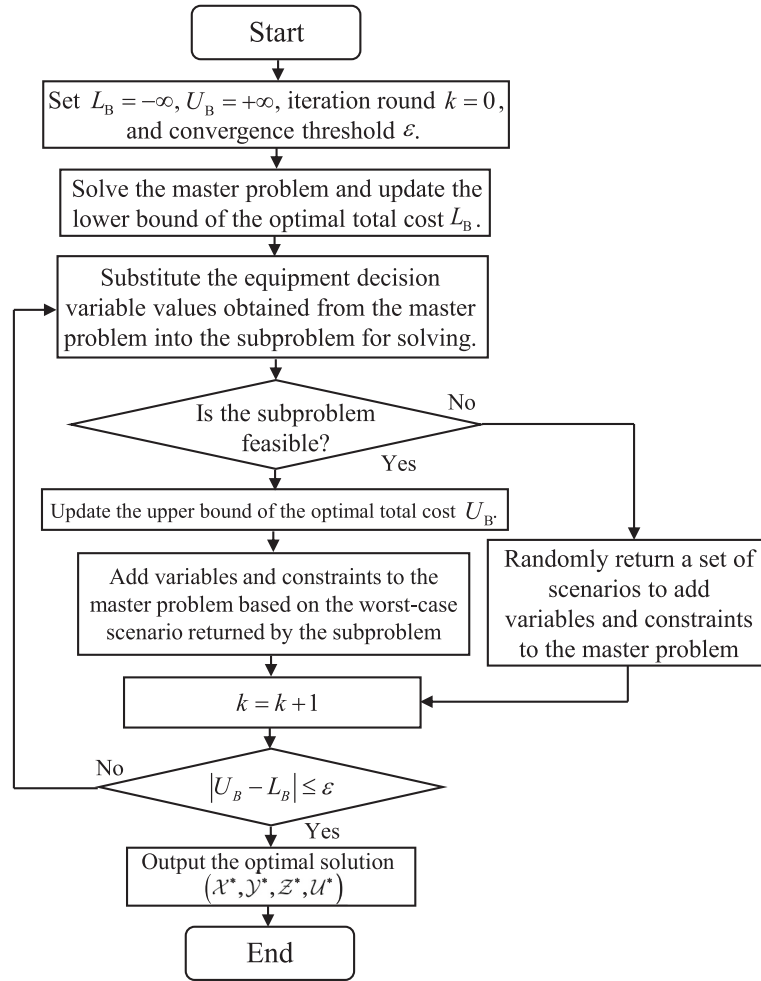


Fig. 6. C&CG algorithm solution process.

4. Case studies

4.1. Data description

The dataset comes from a city in eastern China, covering the period from May 14, 2021, to November 27, 2023, with a 15-min time resolution, resulting in 96-time steps per day. All models were developed using Python and PyTorch, and run on a personal computer equipped with an Intel i7-13,700 5.20 GHz CPU, 16 GB of RAM, and an NVIDIA RTX 4070 GPU. The optimization solver used was GUROBI 11.0, while the power system model was built using PANDAPOWER [23]. The dataset was split into 80 % for training and 20 % for testing to handle uncertainty factors.

This study uses the modified IEEE 37-node system to validate the effectiveness of the proposed method. The equipment parameters of the test system are shown in Table 1. The topology of the coupled power and WDS is illustrated in Fig. 7. The system consists of 18 load nodes, 4 PV power nodes (with capacities of 35 MW, 25 MW, 30 MW, and 30 MW), 4 wind power nodes (with capacities of 50 MW, 40 MW, 55 MW, and 35 MW), 2 battery energy storage nodes, 4 generation unit nodes, and 16 water distribution system nodes. Additionally, nodes 10, 21, and 35 are connected to water tanks, each with a capacity of 2000 m³. The outputs of renewable energy, power demand, and water demand are detailed in Appendix B.

The industrial electricity prices for the power system are presented in Table 2. In the experiment, the confidence levels for wind power generation, photovoltaic generation, and load are set to 85 %, 90 %, and 95

Table 1

Summary of equipment parameters in the test system.

Equipment	Parameters	Values
MT	O&M Cost (¥/MWh)	67.525
	Fuel Cost (¥/MWh)	483.149
	Rated Power (MW)	4×30
	Minimum Output (MW)	4×3
WT	O&M Cost (¥/MWh)	27.463
	Rated Power (MW)	180
PV	O&M Cost (¥/MWh)	8.747
	Rated Power (MW)	120
	O&M Cost (¥/MWh)	28.364
HS	Rated Capacity (MWh)	2×10 (0.5C)
	SOC	0.1–0.9
	Charging/Discharging Efficiency	0.92/0.94

%. Wind power exhibits high fluctuations, so a lower confidence level is adopted to enhance system flexibility. PV generation is relatively stable, thus set at 90 %. Load forecasting is critical for power supply security and cost control, requiring a higher confidence level to ensure system stability.

4.2. The impact of water tanks on system flexibility regulation

4.2.1. Size

This study employs a data-assisted method to generate seasonal scenarios and analyze the impact of different water tank sizes on the upward/downward flexibility requirements of the power system, as

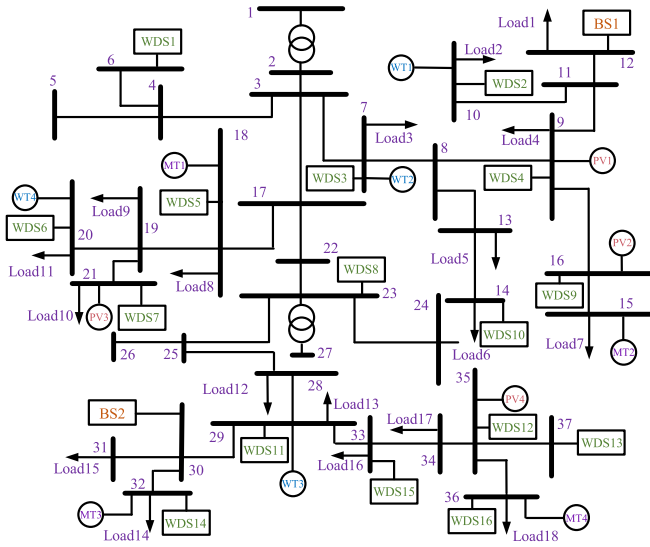


Fig. 7. Modified IEEE 37-node power system structure.

Table 2
Time-of-use tariffs.

States	Time Period (h)	Price (¥/kWh)
Valley Price	0–6	0.34
Mid-Peak Price	6–8 & 22–24	0.78
High Peak Price	8–10 & 12–18 & 20–22	1.20
Peak Price	10–12 & 18–20	1.83

shown in Fig. 8. The short and long dashed lines represent the upward and downward flexibility requirements of the system without water tanks.

After integrating the water distribution system, the system's downward regulation capability is significantly enhanced, while upward regulation capability is also improved. The results indicate that as the water tank size increases, the upward/downward flexibility requirements gradually decrease and eventually stabilize beyond a certain size. Additionally, seasonal variations in renewable energy generation affect water tank requirements. In autumn, when renewable energy generation is relatively low, a smaller water tank capacity is sufficient to meet flexibility needs. However, in winter and spring, when renewable energy generation is higher, the system requires a larger water tank capacity to provide adequate regulation capability.

Overall, the system's demand for downward flexibility is greater than that for upward flexibility. During renewable energy generation peaks, variable-speed pumps operate at high power to store water, significantly enhancing downward flexibility. Conversely, when generation is low, the system relies on stored water pressure and low-speed pumps, reducing the impact of water supply loads on the power grid

and alleviating the upward adjustment pressure on generators.

4.2.2. Location

To maintain analytical consistency, this study selects a 2000 m³ water tank and connects it to different water distribution system nodes to examine the impact of tank location on the system's upward/downward flexibility. The specific results are shown in Table 3. The test system consists of 16 water distribution nodes. Given the large number of nodes, the water tanks are classified into three categories based on their geographical distribution (Fig. 7): Nodes near renewable energy sources; Nodes near conventional generators; Nodes far from any energy generation sources.

As shown in Table 3, placing the water tank at renewable energy nodes significantly improves both upward and downward flexibility, with an improvement range of 68.237 % to 83.187 %. The coordinated operation of pumps and water tanks effectively mitigates renewable energy output fluctuations. When the water tank is located near conventional generators, the flexibility improvement is limited (up to 29.243 %). If the water tank is far from any energy generation source, its regulation capability is negligible due to power flow constraints in the transmission network.

4.3. Algorithm performance analysis

This paper is based on a two-stage robust optimization framework and employs the following methods to generate uncertainty sets for comparative analysis: Bayesian Linear Regression (Method 1) [24], Dirichlet Process (Method 2) [25], Gaussian Process Regression (Method 3) [26], Long Short-Term Memory Network (Method 4) [27], Temporal Convolutional Network (Method 5) [28], Informer (Method 6) [29], and the method proposed in this paper (Method 7). Neural network methods utilize an ensemble approach to generate probability distributions to capture uncertainty.

To compare the performance of each method, interval coverage, average interval width, and a comprehensive index are used to evaluate the prediction performance of the models [30]. In addition, by setting different confidence levels, the risk tolerance in the robust optimization scheduling is adjusted to achieve a balance between robustness and economic efficiency in scheduling.

4.3.1. Uncertainty

Table 4 presents the evaluation results of different methods on the

Table 3
Impact of water tank location on the flexibility regulation of the power system.

WDS	Flexibility Requirement Improvement	
	Upward	Downward
2,3,4,6,7,9,11,12	68.237 % ~ 79.468 %	75.986 % ~ 83.187 %
5,14,16	11.514 % ~ 23.416 %	19.692 % ~ 29.243 %
1,8,10,13,15	3.648 % ~ 9.134 %	6.267 % ~ 13.294 %

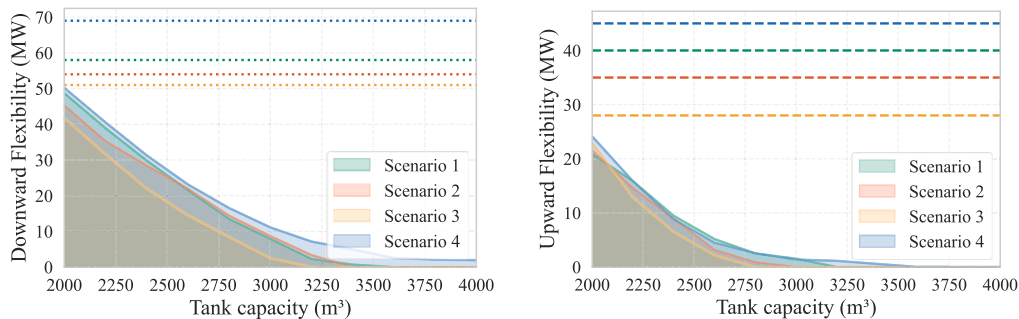


Fig. 8. Impact of water tank size on the flexibility of the power system.

Table 4
Performance analysis of uncertainty.

Confidence level	Method	PV			WT			Load		
		ECP	AW	S_{β}	ECP	AW	S_{β}	ECP	AW	S_{β}
95 %	1	0.936	3.806	4.812	0.916	7.003	8.95	0.965	7.275	8.748
	2	0.974	3.979	5.197	0.923	7.125	8.848	0.93	7.319	8.68
	3	0.969	3.633	4.772	0.989	6.851	8.731	0.938	7.153	8.59
	4	0.943	3.36	4.137	0.974	6.394	7.728	0.96	6.488	7.511
	5	0.957	3.287	4.118	0.927	6.363	7.551	0.955	6.482	7.418
	6	0.948	3.114	3.905	0.971	6.277	7.402	0.947	6.416	7.492
	7	0.951	3.141	3.732	0.965	6.189	7.194	0.95	6.43	7.132
90 %	1	0.873	2.781	3.725	0.859	5.771	7.298	0.88	6.225	7.174
	2	0.871	2.867	3.874	0.853	5.957	7.668	0.925	6.299	7.397
	3	0.879	2.753	3.523	0.871	5.585	6.755	0.889	5.991	6.667
	4	0.878	2.638	3.43	0.869	5.398	6.615	0.885	5.677	6.449
	5	0.882	2.524	3.194	0.874	5.305	6.372	0.889	5.503	6.15
	6	0.895	2.47	2.762	0.913	5.312	6.005	0.905	5.429	5.898
	7	0.903	2.396	2.626	0.908	5.326	5.876	0.897	5.456	5.869
80 %	1	0.827	2.472	3.023	0.853	5.253	6.547	0.836	5.562	6.327
	2	0.834	2.554	3.398	0.845	5.407	6.509	0.832	5.768	6.69
	3	0.832	2.389	3.079	0.851	5.098	5.583	0.835	5.356	6.023
	4	0.82	2.105	2.432	0.767	4.512	5.474	0.778	4.701	5.271
	5	0.822	2.099	2.426	0.772	4.488	5.419	0.781	4.655	5.078
	6	0.821	2.087	2.269	0.774	4.575	5.454	0.824	4.717	5.086
	7	0.814	2.163	2.321	0.822	4.651	5.362	0.789	4.802	5.09

uncertainty set. Overall, the load forecasting scenario performs best, while the wind power forecasting scenario performs the worst. As the confidence level increases, all methods show an increase in interval width (AW) and comprehensive metric (S_{β}) across different scenarios.

From Table 4, it is evident that neural network methods outperform statistical methods in all three forecasting scenarios (photovoltaic (PV), wind power (WT), and load (Load)). Load forecasting achieves the highest empirical coverage probability (ECP), with a maximum value of 96.5 % (Method 1, 95 % confidence level), while maintaining a moderate AW and S_{β} . Wind power forecasting has a lower ECP, while both AW and S_{β} are significantly higher than in other scenarios, indicating wider prediction intervals and overall lower forecasting performance. Photovoltaic forecasting shows intermediate performance, with relatively stable indicators across all methods. In contrast, statistical methods perform worse. This may be because they rely on fixed parameters (e.g., mean, variance) or clustering techniques, which struggle to capture the periodic variations in high-dimensional time-series data, leading to lower prediction accuracy.

Further comparing neural network methods, although the proposed method does not show a clear advantage in interval width (AW) compared to Informer and TCN, and even performs slightly worse in some cases, it achieves better balance across ECP, AW, and S_{β} . This prevents the issue of overly narrow prediction intervals, which could lead to inaccurate estimations and reduced overall performance. Overall, the proposed method demonstrates the best performance across different confidence levels, effectively capturing uncertainty information while ensuring high confidence and an optimized comprehensive metric (S_{β}), thereby improving prediction reliability.

4.3.2. Optimization results

Based on the uncertainty results generated by each model in the previous section, this paper compares the operating costs of different optimization methods under different risk tolerance levels, as shown in

Table 5. Scenario A involves only the power system, while Scenario B integrates the water distribution system, introducing variable-speed pumps and water tanks as flexibility resources. To verify the effectiveness of the two-stage robust optimization (TRO) method proposed in this paper, MILP (Mixed-Integer Linear Programming) is used as the benchmark method (assuming the predicted values are the expected values and ignoring uncertainty), and the stochastic optimization (SO) method (assuming the prediction errors follow a normal distribution and generating the uncertainty set using Method 3).

Table 5 shows that, regardless of the optimization method used, the operating cost in Scenario B is lower than in Scenario A, with the overall cost reduced by approximately 8.68 % to 13.16 %. This indicates that the water distribution system, by providing flexibility resources such as variable-speed pumps and water tanks, effectively reduced the power system's output demand and lowered the operating costs. Among the different methods, MILP has the lowest cost but does not consider the uncertainty of renewable energy output and load demand, which may lead to scheduling failure or increased system safety risks during large fluctuations. The SO method is theoretically superior to TRO, but due to the assumption that prediction errors follow a normal distribution, it struggles to handle the skewness, heavy tails, or asymmetry of renewable energy output and load demand, leading to suboptimal results. In contrast, the data-assisted two-stage robust optimization method proposed in this paper is more accurate in capturing the characteristics of renewable energy and load, improving the precision of the uncertainty set description, reducing conservativeness, and performing better in balancing system economy and safety.

When the risk tolerance is reduced from 20 % to 5 %, the operating costs of each optimization method increase by 9.57 % to 12.81 %. This is primarily because a lower risk tolerance requires the optimization scheduling to account for a larger uncertainty set to handle the extreme fluctuations in renewable energy output and load demand, thus increasing system reserve capacity and scheduling flexibility, which

Table 5
Comparison of operating costs of different optimization methods under different risk tolerances.

Risk Tolerance	MILP		SO		TRO	
	A	B	A	B	A	B
20 %	75,156.12	65,840.78	79,856.49	69,342.76	76,218.95	66,873.43
10 %	80,946.46	71,684.53	84,757.83	75,468.87	82,078.65	73,390.18
5 %	83,884.56	75,164.14	88,548.14	79,537.27	85,690.61	76,310.85

raises the operating costs. In contrast, when the risk tolerance is increased to 20 %, the prediction intervals become more concentrated, reducing the uncertainty disturbance in unit scheduling, and the operating costs decrease. However, at higher risk tolerance levels, if renewable energy or load experiences significant fluctuations, it may pose greater risks to system stability. Therefore, in practical power-water system scheduling, a low risk tolerance (5 %) is more suitable for systems with high renewable penetration and large load fluctuations to ensure safe and reliable operation, while a high-risk tolerance (20 %) is suitable for systems with stable renewable output and sufficient reserve capacity to reduce operating costs.

4.4. Day-ahead dispatching plan

In practical operation, schedulers need to dynamically adjust the risk tolerance α based on the fluctuations in renewable energy output and load changes to balance the system's security and economy. This case study designs four scheduling scenarios to analyze the impact of power-water coupling optimization and dynamic risk tolerance: Scenario A considers only the power system scheduling, with no optimization of the water distribution system and a fixed $\alpha = 5\%$; Scenarios B and C introduce optimization of the water distribution system, with $\alpha = 5\%$ and $\alpha = 20\%$; Scenario D, based on the previous scenarios, dynamically adjusts the risk tolerance α_t according to the safety index B_{eq} , concentration index C_{cen} , and violation index V_{vio} .

The day-ahead scheduling plans for the four scenarios are shown in Fig. 9. In Scenario A, due to the lack of coordinated scheduling with the water distribution system, the system relies on traditional units to cope with renewable energy fluctuations and load changes, leading to frequent power output fluctuations, high reserve capacity demand, and increased operational costs. Scenario B, through the coordinated scheduling of the water distribution and power systems, effectively smooths the load by reducing pump power consumption and alleviating pressure on the power system during high load periods (6:00–12:00, 18:00–22:00), while utilizing surplus renewable energy to store water during low load periods (1:00–5:00, 12:00–18:00). The operating cost is reduced by 10.54 % compared to Scenario A, but due to the lower risk

tolerance, reserve capacity demand remains relatively high. Scenario C adopts a high-risk tolerance strategy with more aggressive scheduling, prioritizing the use of renewable energy and storage to reduce unit output, further lowering operating costs by 16.87 % compared to Scenario B. However, this strategy leads to more frequent adjustments to the schedule during large fluctuations in renewable energy, increasing the risk of scheduling failure and insufficient reserve capacity, especially during high load periods. Scenario D, through dynamic adjustment of the risk tolerance α_t , adapts to optimize scheduling based on the system's state: reducing α_t during high uncertainty periods (6:00–12:00, 18:00–22:00) to enhance system robustness, and increasing α_t during stable periods (1:00–5:00, 12:00–18:00) to reduce reserve costs. Compared to Scenario C, Scenario D results in a 3.56 % higher operating cost, but by flexibly adjusting the risk tolerance, it balances economy and security while further improving the adaptability of reserve capacity allocation and the flexibility of unit scheduling.

Additionally, the coordinated operation of the power-water system effectively smooths the water demand of the water distribution system,

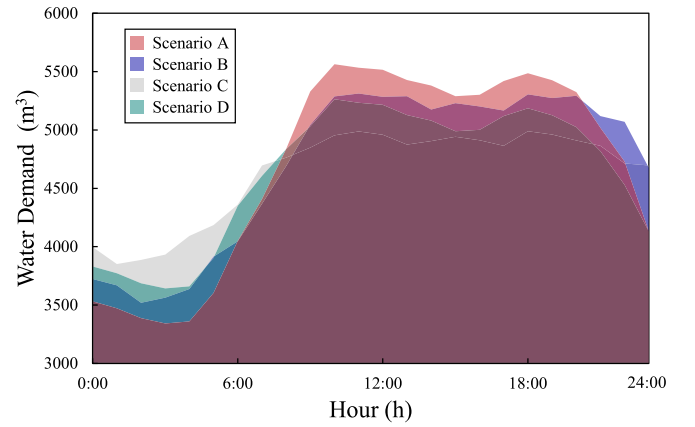


Fig. 10. Water demand of the water distribution system.

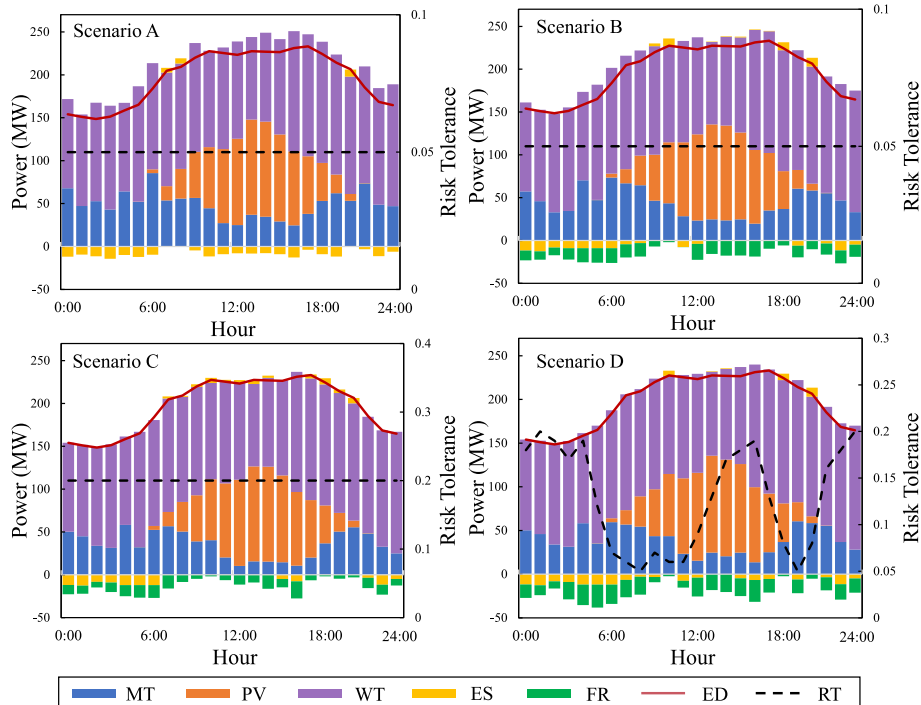


Fig. 9. Day-ahead scheduling results under different scenarios.

altering the water consumption pattern during peak demand periods. As shown in Fig. 10, during peak water consumption periods (6:00–12:00, 16:00–20:00), variable-speed pumps and water tanks, as flexibility resources, consume only the electricity necessary to meet water distribution demand, reducing the pressure on the grid. During low load periods (1:00–5:00, 21:00–24:00), the pumps operate at high speed, utilizing surplus renewable energy to store water, ensuring sufficient water supply during high load periods. Through coordinated operation of the pumps and water tanks, the pumps absorb excess renewable energy and store water in the tanks during low load periods, while reducing power consumption during high load periods by relying on the water tanks to provide water pressure, thus maintaining the normal operation of the water distribution system.

4.5. Analysis of scheduling results

4.5.1. Unit commitment analysis

In power grid scheduling that considers renewable energy uncertainty, the POR of unit commitment describes the boundary of unit output and its coordination relationship. This paper selects three time periods—3:00, 14:00, and 20:00—to visualize unit output under different risk tolerance levels using a two-dimensional representation, as shown in Fig. 11. The red boundary in the figure represents the Safe Operation Region, which indicates the feasible output range of the units under extreme conditions to ensure that safety constraints are not violated.

Compared to Scenario A, the overall unit output in Scenarios B/C decreases, and the POR shrinks, demonstrating that coordinated scheduling with the water distribution system effectively reduces the burden on the power system and improves unit operational stability. In Scenarios B and C, increasing risk tolerance makes the scheduling more concentrated in the economic operation range: Low risk tolerance (light

blue region) provides greater scheduling flexibility but increases reserve capacity demand and operating costs; High risk tolerance (dark blue region) reduces reserve requirements and lowers costs, but renewable energy fluctuations may increase operational risks. This change reflects the trade-off between economy and security.

Scenario D, by adaptively adjusting the risk tolerance, enhances the adaptability of unit commitment. Compared to Scenarios B/C, its POR dynamically changes over time: During high-uncertainty periods (e.g., 20:00), risk tolerance is lowered, expanding the operating range and enhancing system robustness; During stable periods (e.g., 3:00), risk tolerance is increased, making scheduling more concentrated on economic operation, reducing reserve capacity demand, and improving output stability. Compared to fixed risk tolerance strategies, Scenario D balances economy and security across different time periods, lowering reserve costs while mitigating operational risks.

4.5.2. Evaluation index analysis

To quantitatively assess the operational performance of different scheduling scenarios, this paper calculates the safety index B_{eq} , the concentration index C_{cen} , and the violation index V_{vio} at 3:00, 14:00, and 20:00, with the specific values for each scenario shown in Fig. 12.

During the nighttime stable period (3:00), B_{eq} in Scenario A does not significantly differ from the other scenarios, indicating that when load fluctuations are small, different scheduling strategies have a relatively minor impact on system security. However, during high-uncertainty periods (14:00 and 20:00), B_{eq} in Scenarios B/C is significantly higher than in Scenario A, demonstrating that coordinated scheduling of the water distribution system effectively mitigates renewable energy fluctuations. At the same time, Scenario A exhibits a relatively high C_{cen} , as the lack of coordination with the water distribution system forces units to handle larger output fluctuations, leading to a more dispersed overall scheduling distribution. Additionally, as risk tolerance increases, B_{eq}

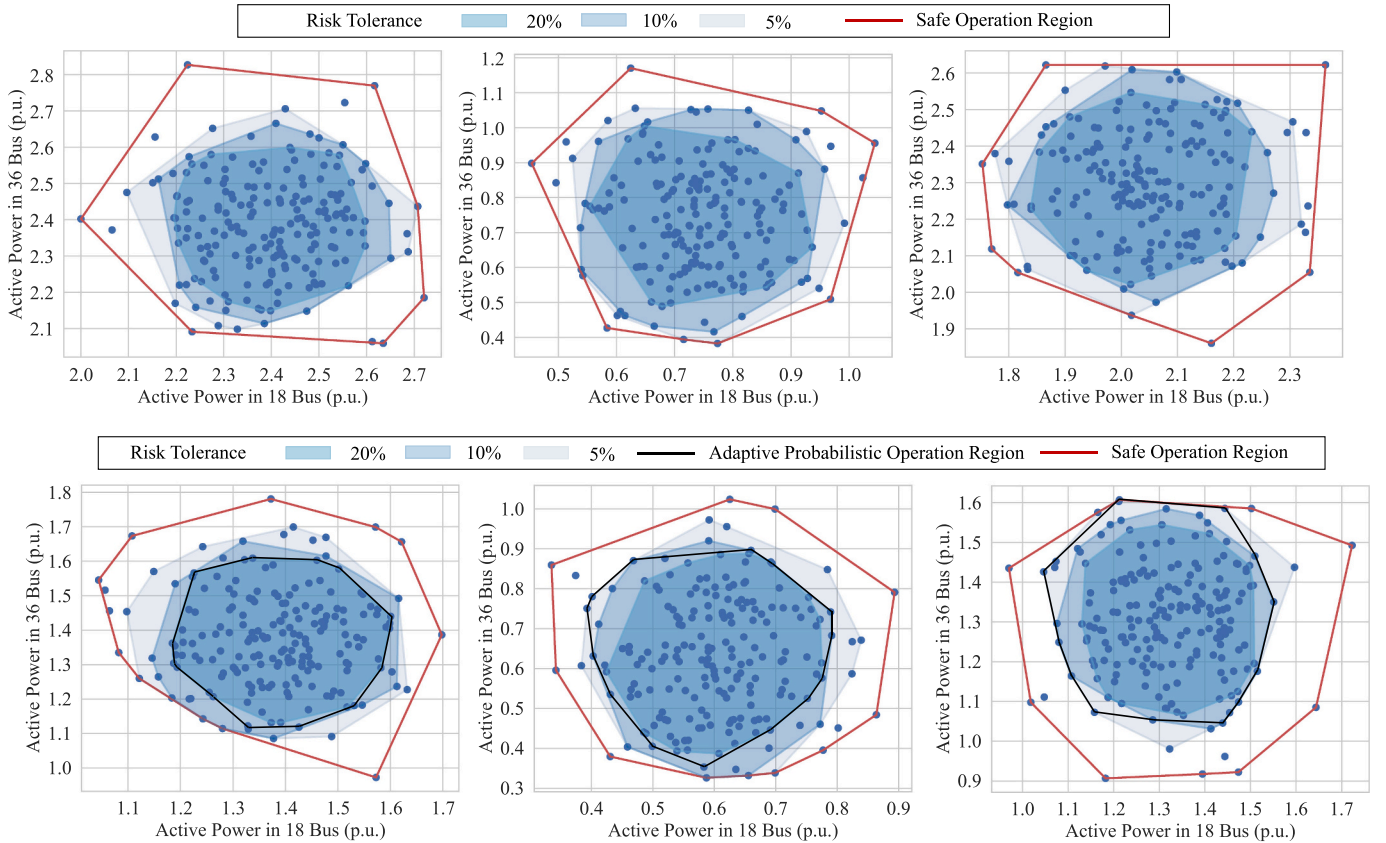


Fig. 11. Probabilistic operation region of unit commitment.

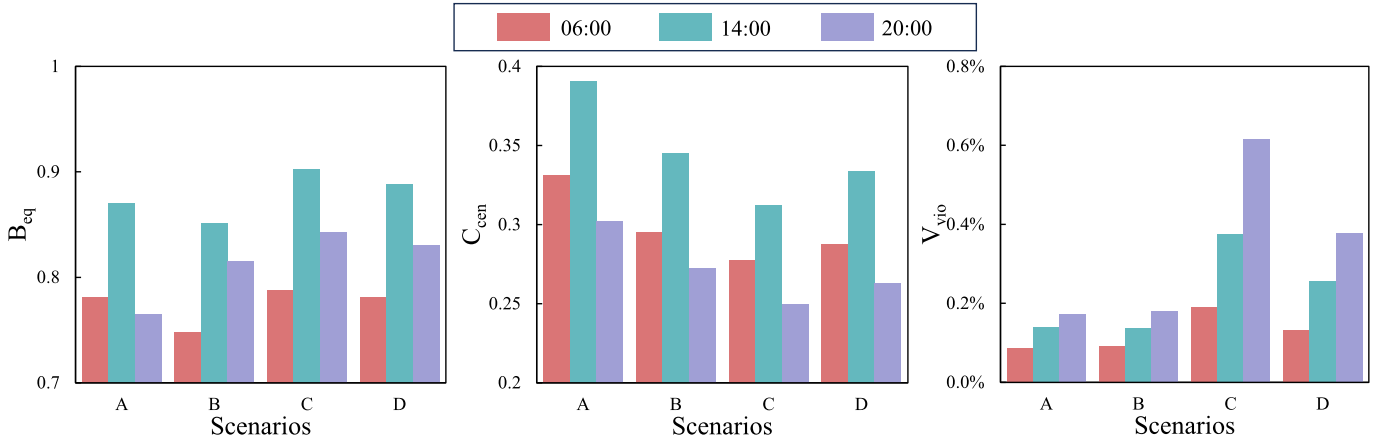


Fig. 12. Evaluation indices of the unit probabilistic operation region.

gradually increases, C_{cen} decreases, indicating that scheduling flexibility is reduced, and the system tends to prioritize economic optimization while reducing reserve requirements. However, V_{vio} gradually increases, suggesting that under high risk tolerance, the unit operation region becomes more constrained, and renewable energy uncertainty may lead to operational violations in certain periods.

Scenario D, which dynamically adjusts risk tolerance, optimizes its indices adaptively across different time periods, performing between Scenarios B and C at different risk tolerance levels. During high-uncertainty periods (20:00), Scenario D moderately reduces B_{eq} and increases C_{cen} , improving system robustness while addressing the issue of excessive V_{vio} in Scenario C at high risk tolerance, ensuring that safety constraints are not violated. During stable periods (3:00), B_{eq} increases moderately while C_{cen} decreases, reducing reserve requirements and making scheduling more economical. This proves that the dynamic adjustment strategy effectively balances security and economic performance, enabling real-time adaptation between robustness and cost optimization.

It is important to note that under the same risk tolerance, multiple unit operating points may satisfy safety constraints but exhibit different output distributions. When making scheduling decisions, priority should be given to schemes with higher B_{eq} (improved security), lower C_{cen} (better concentration in scheduling), lower V_{vio} (reduced violations of constraints), to ensure the stability and cost-effectiveness of the system.

5. Conclusion

This paper proposes a data-assisted two-stage robust optimization scheduling method for the coordinated operation of urban water distribution and power grid integrated energy systems and constructs a probabilistic operation region to improve system security and economy. The main conclusions are as follows:

1. Quantification of the impact of flexibility resources on the power system: The demand for downward flexibility from the water distribution system is higher than the upward demand. Moreover, when the water tank is located near the renewable energy generation node, the system's flexibility improvement is more significant. This indicates that proper water resource scheduling helps reduce the impact of renewable energy on the power system.
2. The proposed data-assisted method effectively improves optimization results: This method can effectively narrow the forecasting

interval while ensuring the coverage of operational points, yielding higher comprehensive performance metrics. By incorporating the probability distribution information from historical data, the optimization approach can better cope with uncertainty factors and enhance scheduling robustness.

3. Probabilistic operation region and evaluation indices enrich scheduling diversity: Operators can adjust scheduling strategies in different scenarios to dynamically balance security and economy. The introduction of the dynamic risk tolerance mechanism allows the system to adjust risk tolerance in real-time based on factors such as renewable energy forecast accuracy and load fluctuations, preventing the security or economic losses caused by a fixed tolerance level.

Future research will focus on developing systematic mathematical models to quantify the risk-economy trade-off boundaries and optimize the optimal decision points under different scheduling strategies. Additionally, robust game theory can be used to analyze the risk-economy evolution under different market environments and policy incentives. Furthermore, research can explore emergency response mechanisms for situations where renewable energy exceeds the probabilistic operating region, optimizing reserve power deployment and demand-side management strategies to enhance the practical feasibility of scheduling solutions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Manguo Zhou: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization. **Kaiming Zhang:** Writing – review & editing, Investigation, Formal analysis. **Xiangwan Liu:** Investigation, Data curation. **Fan Ma:** Funding acquisition, Data curation.

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Appendix A

Under the uncertainty set $\mathcal{Z}^R$, if the feasible region $\mathcal{Y}^R$ of the grid dispatch plan is a convex set, then all solutions within $\mathcal{Y}^R$ are also feasible

solutions. The proof is as follows

1. Proof by contradiction: Assume there exists $y \in \Omega^R$ such that $y \notin \mathcal{J}^R$. In other words, there is at least one solution within Ω^R but outside $\mathcal{J}^R$, which implies $\Omega^R \setminus \mathcal{J}^R \neq \emptyset$.

2. Relation between POR and the feasible region: Since $\mathcal{S}^R \subseteq \mathcal{J}^R$ (all dispatch plans that satisfy probabilistic constraints are feasible), and Ω^R is the convex hull of $\mathcal{S}^R$, meaning $\mathcal{S}^R \subseteq \Omega^R$, it follows that $\mathcal{S}^R \subseteq \Omega^R \cap \mathcal{J}^R$.

3. Convexity: Since $\mathcal{J}^R$ is a convex set, and Ω^R is also a convex set (as the convex hull of $\mathcal{S}^R$), their intersection $\Omega^R \cap \mathcal{J}^R$ must also be a convex set.

4. Contradiction: By the initial assumption, $\Omega^R \setminus \mathcal{J}^R \neq \emptyset$, meaning that Ω^R is not fully contained within $\mathcal{J}^R$. However, since $\Omega^R \cap \mathcal{J}^R$ is a convex set that contains $\mathcal{S}^R$, if Ω^R contains infeasible solutions, the intersection $\Omega^R \cap \mathcal{J}^R$ would form a convex set smaller than Ω^R , yet this smaller convex set would still cover $\mathcal{S}^R$. This contradicts the fact that Ω^R is the smallest convex hull of $\mathcal{S}^R$.

Thus, the assumption is false, and it is proven that all solutions within Ω^R are feasible solutions.

Appendix B

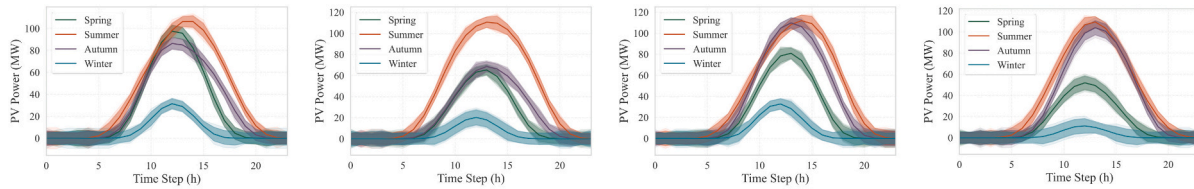


Fig. B1. PV power forecast result.

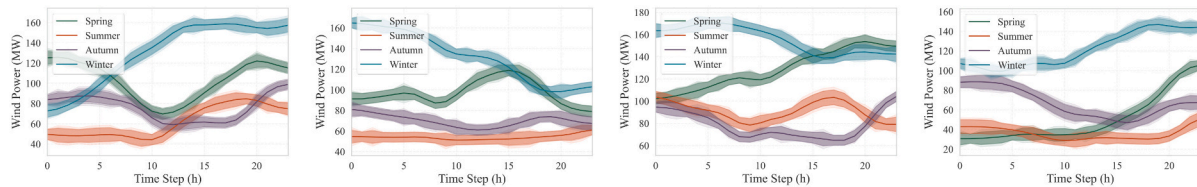


Fig. B2. Wind power forecast result.

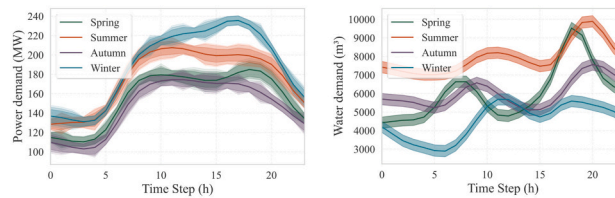


Fig. B3. Electricity load and water demand.

Data availability

The authors do not have permission to share data.

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